DATA ANALYTICS PROJECT

Customer Shopping Behavior Analysis

An analysis of shopping transactions, customer segments, product preferences, and subscription behavior.

3,900	**18**	**37**
Purchases	Columns	Missing review ratings

PROJECT OVERVIEW

This project analyzes customer shopping behavior using transactional data from 3,900 purchases across various product categories. The goal is to uncover insights into spending patterns, customer segments, product preferences, and subscription behavior to guide strategic business decisions.

TOOLS USED

Python • pandas • PostgreSQL • SQL • Power BI

01 | Dataset & Preparation

Dataset Summary

Dataset detail	Description
Rows	3,900
Columns	18
Customer demographics	Age, Gender, Location, Subscription Status
Purchase details	Item Purchased, Category, Purchase Amount, Season, Size, Color
Shopping behavior	Discount Applied, Promo Code Used, Previous Purchases, Frequency of Purchases, Review Rating, Shipping Type
Missing data	37 values in Review Rating column

Exploratory Data Analysis using Python

We began with data preparation and cleaning in Python:

- **Data Loading:** Imported the dataset using `pandas`.
- **Initial Exploration:** Used `df.info()` to check structure and `.describe()` for summary statistics.
- **Missing Data Handling:** Checked for null values and imputed missing values in the Review Rating column using the median rating of each product category.
- **Column Standardization:** Renamed columns to snake case for better readability and documentation.
- **Feature Engineering:** Created `age_group` column by binning customer ages; created `purchase_frequency_days` column from purchase data.
- **Data Consistency Check:** Verified if `discount_applied` and `promo_code_used` were redundant; dropped `promo_code_used`.
- **Database Integration:** Connected Python script to PostgreSQL and loaded the cleaned DataFrame into the database for SQL analysis.

02 | SQL Business Analysis

Structured analysis was performed in PostgreSQL to answer key business questions.

1. Revenue by Gender

Compared total revenue generated by male vs. female customers.

Gender	Revenue
Female	75,191
Male	157,890

2. High-Spending Discount Users

Identified customers who used discounts but still spent above the average purchase amount.

Customer ID	Purchase Amount
2	64
3	73
4	90
7	85
9	97
12	68
13	72
16	81
20	90
22	62

The SQL result contained 839 rows in total.

3. Top 5 Products by Rating

Found products with the highest average review ratings.

Item Purchased	Average Product Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

4. Shipping Type Comparison

Compared average purchase amounts between Standard and Express shipping.

Shipping Type	Average Purchase Amount
Standard	58.46
Express	60.48

5. Subscribers vs. Non-Subscribers

Compared average spend and total revenue across subscription status.

Subscription Status	Total Customers	Average Spend	Total Revenue
Yes	1,053	59.49	62,645.00
No	2,847	59.87	170,436.00

03 | Customer & Product Insights

6. Discount-Dependent Products

Identified 5 products with the highest percentage of discounted purchases.

Item Purchased	Discount Rate
Hat	50.00
Sneakers	49.66
Coat	49.07
Sweater	48.17
Pants	47.37

7. Customer Segmentation

Classified customers into New, Returning, and Loyal segments based on purchase history.

Customer Segment	Number of Customers
Loyal	3,116
New	83
Returning	701

8. Top 3 Products per Category

Listed the most purchased products within each category.

Rank	Category	Item Purchased	Total Orders
1	Accessories	Jewelry	171
2	Accessories	Sunglasses	161
3	Accessories	Belt	161
1	Clothing	Blouse	171
2	Clothing	Pants	171
3	Clothing	Shirt	169
1	Footwear	Sandals	160
2	Footwear	Shoes	150
3	Footwear	Sneakers	145
1	Outerwear	Jacket	163
2	Outerwear	Coat	161

9. Repeat Buyers & Subscriptions

Checked whether customers with >5 purchases are more likely to subscribe.

Subscription Status	Repeat Buyers
No	2,518
Yes	958

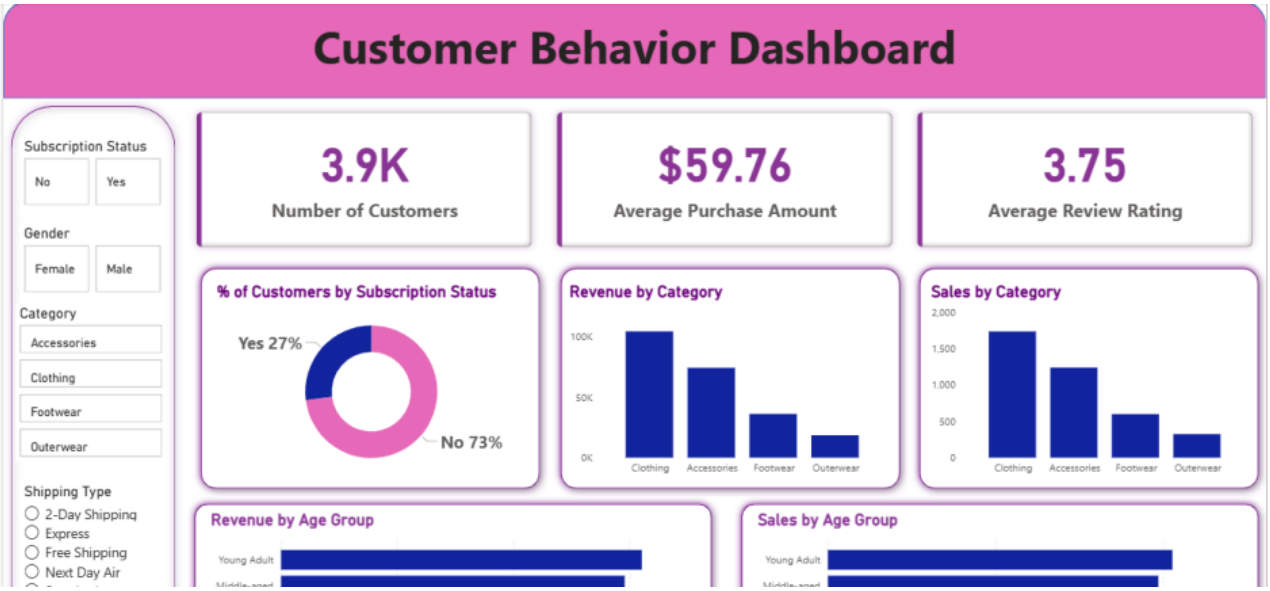
10. Revenue by Age Group

Calculated total revenue contribution of each age group.

Age Group	Total Revenue
Young Adult	62,143
Middle-aged	59,197
Adult	55,978
Senior	55,763

04 | Power BI Dashboard

Finally, we built an interactive dashboard in Power BI to present insights visually.



05 | Business Recommendations

Boost Subscriptions

Promote exclusive benefits for subscribers.

Customer Loyalty Programs

Reward repeat buyers to move them into the “Loyal” segment.

Review Discount Policy

Balance sales boosts with margin control.

Product Positioning

Highlight top-rated and best-selling products in campaigns.

Targeted Marketing

Focus efforts on high-revenue age groups and express-shipping users.

End of Report